

# Quantum 2 HW5

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*Technion*

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## 1 The Aharonov-Bohm Effect

### 1.1 Hamiltonian

Since the magnetic flux is only through the middle of the ring, there is no magnetic field (only a potential) on the ring itself, therefore the hamiltonian won't include an element for the interaction of the spin and the magnetic field. Therefore

$$H = \frac{1}{2m} (\mathbf{p} - e\mathbf{A})^2 \quad (1)$$

inserting the vector potential we have

$$H = \frac{1}{2m} \left( \mathbf{p} - \frac{e\Phi}{L} \right)^2 \quad (2)$$

we can constrain the momentum element because we know the particle is on the ring, therefore

$$\mathbf{p} = -\frac{\hbar}{i} \partial_x = -\frac{\hbar}{iR} \partial_\theta \quad (3)$$

so the hamiltonian is

$$H = \frac{\hbar^2}{2mR^2} \left( -i\partial_\theta + \frac{e\Phi}{2\pi\hbar} \right)^2 \quad (4)$$

and the Schrodinger equation is

$$i\hbar\partial_t\psi = \frac{\hbar^2}{2mR^2} \left( -i\partial_\theta + \frac{e\Phi}{2\pi\hbar} \right)^2 \psi \quad (5)$$

### 1.2 Setting X

I already did this in the previous subsection.  $x$  is defined in  $[0, L]$ . We can define the basic flux unit  $\Phi_0 = 2\pi\hbar/e$  and find

$$H = \frac{\hbar^2}{2mR^2} \left( -i\partial_\theta + \frac{\Phi}{\Phi_0} \right)^2 \quad (6)$$

### 1.3 Energy Spectrum

We'll perform a gauge transform, defining  $\varphi = \frac{\Phi}{\Phi_0}$

$$\psi(\theta) = e^{-i\varphi\theta} \chi(\theta) \quad (7)$$

Which gives us

$$(-i\partial_\theta + \varphi) e^{-i\varphi\theta} \chi(\theta) = (-\varphi\chi(\theta) - i\chi'(\theta) + \varphi\chi(\theta)) e^{-i\varphi\theta} \quad (8)$$

$$= -ie^{-i\varphi\theta} \chi'(\theta) \quad (9)$$

therefore

$$(-i\partial_\theta + \varphi) \psi(\theta) = -ie^{-i\varphi\theta} \partial_\theta \chi(\theta) \quad (10)$$

if we apply the operator again

$$(-i\partial_\theta + \varphi)^2 \psi(\theta) = e^{-i\varphi\theta} (-i\partial_\theta)^2 \chi(\theta) \quad (11)$$

so the Schrodinger equation becomes

$$\frac{\hbar^2}{2mR^2} (-i\partial_\theta)^2 \chi(\theta) = E\chi(\theta) \quad (12)$$

which is the Schrodinger equation for a free particle. This also means  $\chi$  has a solution of the form

$$\chi(\theta) = e^{ik\theta} \quad (13)$$

Next we consider the boundary conditions which changed due to the gauge transform. We know the problem has periodic boundary conditions, therefore

$$e^{-i\varphi\theta} \chi(\theta) = e^{-i\varphi(\theta+2\pi)} \chi(\theta+2\pi) \quad (14)$$

rearranging

$$\chi(\theta) = e^{-2\pi i\varphi} \chi(\theta+2\pi) \quad (15)$$

inserting the solution ansatz

$$e^{ik\theta} = e^{-2\pi i\varphi} e^{ik(\theta+2\pi)} \implies e^{2\pi i\varphi} = e^{2\pi ik} \quad (16)$$

therefore

$$k = n + \varphi \quad n \in \mathbb{N} \quad (17)$$

and this gives us the energy spectrum

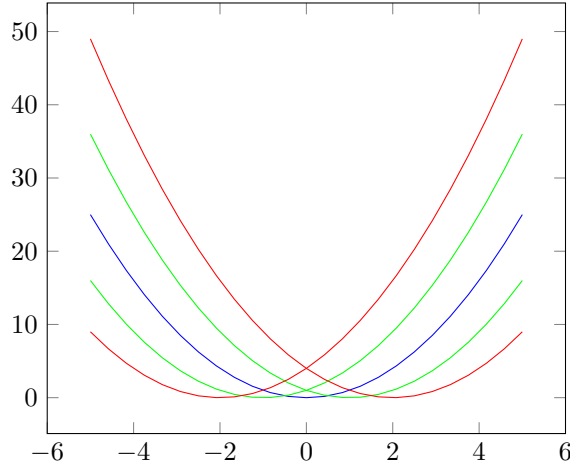
$$E_n = \frac{\hbar^2}{2mR^2} (n + \varphi)^2 \quad (18)$$

as needed.

## 1.4 Gauge transform

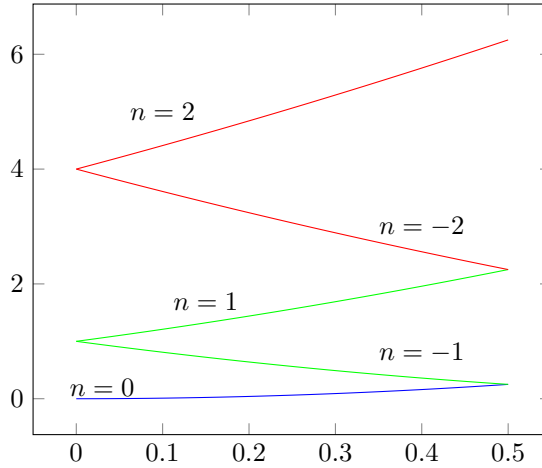
I already showed the energy spectrum in the previous subsection.

## 1.5 Drawing the Spectrum



**Figure 1** – The energy spectrum described by equation 18 for  $n = 0, \pm 1, \pm 2$  marked by blue, green, and red respectively, in multiples of  $\frac{\hbar^2}{2mR^2}$ . I drew this figure using tikz.

## 1.6 Reduced Representation



**Figure 2** – The energy spectrum described by equation 18 for  $n = 0, \pm 1, \pm 2$  marked by blue, green, and red respectively, in multiples of  $\frac{\hbar^2}{2mR^2}$ , in the domain  $0 - 0.5$ . I drew this figure using tikz.

The reason we can reduce our focus on just  $0 - 0.5$  is that the energy spectrum is periodic in terms of  $\varphi$ . Since it depends on two numbers, one continuous ( $\varphi$ ) and one discrete ( $n$ ), any value of  $\varphi$  over  $0.5$  is the same as taking a  $\varphi$  less than one half and adding a negative  $n$ .

## 1.7 Odd and Even Energy Spectrum

We'll define

$$n(p) = \begin{cases} \frac{p}{2} & p \text{ is even} \\ -\frac{p+1}{2} & p \text{ is odd} \end{cases} \quad (19)$$

and insert it into equation 18

$$E_p = \frac{\hbar^2}{2mR^2} \begin{cases} \left(\frac{p}{2} + \varphi\right)^2 & p \text{ is even} \\ \left(-\frac{p+1}{2} + \varphi\right)^2 & p \text{ is odd} \end{cases} \quad (20)$$

but we can take out the minus sign from the parentheses and have

$$E_p = \frac{\hbar^2}{2mR^2} \begin{cases} \left(\frac{p}{2} + \varphi\right)^2 & p \text{ is even} \\ \left(\frac{p+1}{2} - \varphi\right)^2 & p \text{ is odd} \end{cases} \quad (21)$$

because  $-1^2 = 1$ .

## 1.8 Degeneracy

### 1.8.1 For $\varphi = 0$

The energy level for  $p = 0$  will have a degeneracy of 1, the rest will have a degeneracy of 2.

### 1.8.2 For $\varphi = \frac{1}{2}$

If we insert  $\varphi = \frac{1}{2}$  into the energy levels we find

$$E_p = \frac{\hbar^2}{2mR^2} \begin{cases} \left(\frac{p+1}{2}\right)^2 & p \text{ is even} \\ \left(\frac{p}{2}\right)^2 & p \text{ is odd} \end{cases} \quad (22)$$

which is identical to the case  $\varphi = 0$

### 1.8.3 For $0 < \varphi < \frac{1}{2}$

All energy levels will have a degeneracy of 1, because no two additions of  $p$  and  $\varphi$  will match for different  $p$ s.

## 2 Positronium

### 2.1 Reduced Mass

by definition

$$\mu = \frac{m^2}{2m} = \frac{m}{2} \quad (23)$$

### 2.2 Central Force Hamiltonian

The hamiltonian is given by the center of mass momentum element and coulomb element

$$H = \frac{p^2}{2\mu} - \frac{q^2}{4\pi\epsilon_0 r} \quad (24)$$

### 2.3 Energy Levels and Degeneracy

The hamiltonian is structurally the same as that of the hydrogen atom with the replacement of the mass so the spectrum is

$$E_n = -\frac{\mu c^2 \alpha^2}{2n^2} = -\frac{mc^2 \alpha^2}{4n^2} \quad (25)$$

which is half that of the regular hydrogen atom, and the degeneracy is the same as the hydrogen atom,  $n^2$ .

### 2.4 Bohr Radius

The Bohr Radius is

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{e^2 \mu} = \frac{2\pi\epsilon_0 \hbar^2}{e^2 m} \quad (26)$$

which is double that of the hydrogen atom.

### 2.5 Normalized Ground State

The normalized wavefunction in position representation, given by spherical harmonics and Laguerre polynomials, inserting  $n = 1, l = 0, m = 0$  is

$$\psi_{100}(\mathbf{r}) = \sqrt{\frac{4}{a_0^3}} e^{-r/a_0} \cdot 1 \cdot \frac{1}{2} \sqrt{\frac{1}{\pi}} = \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0} \quad (27)$$

inserting the bohr radius we found

$$|\psi_{100}(\mathbf{r})|^2 = \frac{1}{\pi \left( \frac{2\pi\epsilon_0 \hbar^2}{e^2 m} \right)^3} e^{-2r / \left( \frac{2\pi\epsilon_0 \hbar^2}{e^2 m} \right)} \quad (28)$$

I don't see why the question asks about  $\alpha$ , it doesn't depend on the mass, so it doesn't change.

### 2.6 Degeneracy of Orbital Ground State

Each of the particles in the atom has 2 possible spin states, so the total number of states is 4, which is the degeneracy.

## 2.7 Gyromagnetic Ratio

The gyromagnetic ratio is defined as

$$\gamma = \frac{q}{m} \quad (29)$$

both particles' mass is the same but they have opposite charge, so  $\gamma_1 = -\gamma_2$ .

## 2.8 Eigenstates And Eigenvalues

We can rewrite

$$H_{SS} = \frac{A}{\hbar^2} \mathbf{S} \cdot \mathbf{S} = \frac{A}{2\hbar^2} (S^2 - S_1^2 - S_2^2) \quad (30)$$

therefore it is diagonal in the coupled basis and the eigenstates are the triplet and singlet states

$$|S = 1, M = 1\rangle = |++\rangle \quad (31)$$

$$|S = 1, M = 0\rangle = \frac{1}{\sqrt{2}} (|+-\rangle + |-+\rangle) \quad (32)$$

$$|S = 1, M = -1\rangle = |--\rangle \quad (33)$$

$$|S = 0, M = 0\rangle = \frac{1}{\sqrt{2}} (|+-\rangle - |-+\rangle) \quad (34)$$

and the eigenvalues are

$$H_{SS} |S = 1, M = 1\rangle = \frac{A}{2} \left( 2 - \frac{3}{4} - \frac{3}{4} \right) |S = 1, M = 1\rangle = \frac{A}{4} |S = 1, M = 1\rangle \quad (35)$$

but since this is true for any  $M$ ,

$$H_{SS} |S = 1, M\rangle = \frac{A}{4} |S = 1, M\rangle \quad (36)$$

and

$$H_{SS} |S = 0, M = 0\rangle = \frac{A}{2} \left( 0 - \frac{3}{4} - \frac{3}{4} \right) |S = 0, M = 0\rangle = -\frac{3A}{4} |S = 0, M = 0\rangle \quad (37)$$

## 2.9 Difference In Hyperfine Energy

The difference in energy between the two spin states is exactly  $A$ , which itself is proportional to the multiple of the gyromagnetic ratios of both particles and the reduced mass to the 6th power ( $a_0^3$  and it contains  $\mu^2$ )

$$\Delta E \propto \gamma_1 \gamma_2 \mu^6 \quad (38)$$

$\gamma_1$  is the electron's gyromagnetic ratio instead of the protons, which is 3 orders of magnitude larger. The reduced mass is half the electron's mass. If we sum both contributions we find that the energy difference is larger by 2 orders of magnitude, which is the difference in frequencies.

## 2.10 A in terms of constants

We know that

$$A = -\frac{2}{3} \frac{\hbar^2}{\varepsilon_0 c^2} \gamma_1 \gamma_2 |\psi_{100}(0)|^2 \quad (39)$$

if we insert the gyromagnetic ratios and the wavefunction

$$A = -\frac{2}{3} \frac{\hbar^2}{\varepsilon_0 c^2} \frac{e^2}{m^2} \frac{1}{\pi a_0^3} = -\frac{2\hbar^2 e^2}{3\pi \varepsilon_0 m^2 c^2} \frac{1}{\left(\frac{2\pi \varepsilon_0 \hbar^2}{e^2 m}\right)^3} \quad (40)$$

$$= -\frac{e^8 m}{12\pi^4 \varepsilon_0^4 c^2 \hbar^4} = \frac{1}{3} m c^2 \alpha^4 \quad (41)$$

inserting the numbers we find

$$A = \frac{1}{3} \cdot 0.511 \text{ MeV} \cdot \frac{1}{137^4} = 4.8 \times 10^{-4} \text{ eV} \quad (42)$$

## 2.11 Hyperfine Transition Frequency

The hyperfine transition frequency is defined as

$$f = \frac{\Delta E}{h} = \frac{A}{h} = 116.1 \text{ GHz} \quad (43)$$

## 2.12 Annihilation Energies

Since the annihilation element doesn't affect the states for  $S = 0$  the energy remains the same

$$E_{S=0} = -\frac{3}{4} A \quad (44)$$

but the energy for  $S = 1$  changes to

$$E_{S=1} = \frac{1+3}{4} A = A \quad (45)$$

## 2.13 Annihilation hyperfine frequency

same as we did before

$$f = \frac{\Delta E}{h} = \frac{7A}{4h} = 203.1 \text{ GHz} \quad (46)$$